

Call with [REDACTED] – April 5, 2018 -

PhD in molecular biology from OH University – worked on genome transcription

Worked as Fellow at Walter Reed Institute of Research (not the WR Hospital)

Worked at National Cancer Institute in AIDS and Malignancy branch.

Was offered position in “dark side”

- Moved into genetics and genomics. Went from AI analysis (software side) to Iceland working on external scientific collaborations. In charge of largest genotyping facility.
- ?????Sequencing of Atacama genome was Illumina (at the time there were only 4 sequencing techs total)
- ?????Bougie next code company – he led this company

-RL knows everyone in the field.

He approached Garry shortly after the release of “Sirius” film and offered his help. At the time he headed company called Complete Genomics nearby. The technology they had at Complete Genomics was best re sequencing and most appropriate to what needed to be done. Offered to help.

Emphasized that it was a very strange call.

- “Im a believer but this thing is human” – quote from [REDACTED]
 - o When guy called Nolan
 - o Nolan told him that 70-80% of sequence reads mapped (was saying publicly that 90% of sequences mapped)
 - Can’t know what it is based on 70-80% mapping
 - o Date of the call was in 2013
 - o Asked Nolan if there was any DNA left over to sequencing but Nolan was closed off to this idea
 - o RL thinks that [REDACTED] wasn’t the right person to sequence the data
 - o Nolan pulled in 2 guys that carry a lot of weight to help with the sequencing – Bustamante and Butte (?)
 - o Even if it was high quality DNA (which is not) Nolan makes critical biased assumption that its human
 - Sequencing has 3 basic parts: 1) primary 2) secondary 3) tertiary
 - Primary = taking a molecule and turning it into digital information
 - Secondary = raw information and using bioinformatics and algorithms and try to make it look like a genome
 - o Nolan used bill 37 of the human genome to map the Atacama DNA
 - Requires high quality DNA, but Nolan used low quality DNA
 - Nolan didn’t provide enough information about mitochondria
 - Tertiary = variance in genome and trying to interpret and running software
 - Nolan did something wrong: when you get molecules that are digitized they are raw reads. Nolan was mapping reads to a human scaffold. He was trying to fit

the reads to a human, assuming it was human (wrong). They through out 360,000 variants. Normal human being has 200 variants? Nolan should've used a **reference read assembly**

- RL said that Nolan is a nobody in Genetics field.

RL said that he has been involved in genetics from beginning of any kind of advanced technology used with it.

SMG said Nolan's paper was "Complete Rubbish". RL said, "It is. It is."

RL said its "laughable what Nolan says."

"Even if it were perfect quality DNA, it is wrong from the get go. " Nolan assumed that it was human.

Nolan should have worked with "raw reads" but instead was matching "reads" to human DNA.

Should have done the matching "reference free assembly" so would have been using raw data and not referencing against human DNA.

363,000 variants were thrown out by Nolan. A normal human has 200 variants.

He went 11x deep but should have gone several 1000

RL – the quality of the DNA sequencing was "dog shit".

- [REDACTED] and [REDACTED] are only credible people on the paper
- [REDACTED] used GA2 – did not use state of the art sequencing technology
- GA2 was old even then "very low quality". They did not use their best stuff. "Baffling" that they did not use their best stuff.

RL: "Why did he bother to write a paper at all."

When SMG mentioned \$3.2 million DOD grant – RL just said "Oh" with implication that he understood right away that Nolan was bought.

RL: "They didn't do the right work. How did they even know it had sex chromosomes."

- Between generations there are about 6-7 genetic mutations that accrue
 - o The kid wouldn't have been able to develop with this genetic load mutation it would be incompatible with life
 - o Fetus shouldn't have divided out of the embryo stage
 - o Nolan missed the fundamentals.
 - o Nolan assumed too much – how do we know that this thing supports sexually?
 - Assumed there was an accrual of mutations between the specimen and the previous generation, which is based on false assumptions

RL was asked by SMG why Bustamante and Butte would have been involved. "Cronyism". They would not contradict him.

Genomics Research is a very poor journal. Journals of stature in this field would be Science, Nature, Cell

Nolan's attitude that it was human from the original conversation is what had RL worried:

RL : "This kind of stuff is totally pervasive . This kind of corruption is why I left academics."

RL now lives in Michigan, but on airplanes most of the time. VERY reluctant to write anything and DOES NOT want his name used.